Rebecca Kong

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EDUCATION

HARVEY MUDD COLLEGE **Claremont, CA**

B.S. Engineering, Expected May 2028

GPA: **3.94/4.00** | Harvey Mudd Merit Scholarship | Dean's List | Tau Beta Phi Engineering Honor Society

Relevant Coursework: Digital Electrical & Computer Engineering, Experimental Engineering, Engineering Systems, Intro to Engineering Design & Manufacturing, Principles of Computer Science, Applied Math for Engineering, Electric and Magnetic Circuits/Devices, Microprocessor System: Design & Applications, Advanced Systems Engineering

SKILLS

Technical: Python, Java, JavaScript, MATLAB, SystemVerilog, YOLOv8, OpenCV, CAD

Languages: English (fluent), Mandarin Chinese (conversational)

EXPERIENCE

Embedded System R&D Contractor, Blackrock Neurotech *Aug 2026 - Present*

Industry-sponsored project to deliver a wireless data transmission device for a BCI

- Designing and implementing a low power, high data transfer rate wireless device for collecting brain signal data for training purposes
- Ongoing effort for profiling data throughput with different ESP32 MCUs to integrate wireless connection for the data transmission process

AI Product Analysis Intern, SENSETIME - Beijing, China *Summer 2026*

Product development researcher for UniRead (an AI-based English learning system)

- Performed competitive analysis for English-learning platforms to identify gaps in content discovery and community experiences
- Analyzed customer needs against platform limitations and operational requirements to proposed AI specific feature requirements
- Collaborated in a Mandarin-speaking work environment, iterating on product requirements and feature proposals based on team feedback.

Student Researcher, Entrepreneurship Studio Program, Harvey Mudd College - Claremont, CA *Summer 2025*

Computer-vision tennis analysis platform using object detection and automated match processing

- Developed an AI-based application PointView, which utilized computer vision and machine learning to analyze tennis matches by tracking player and ball with YOLOv8.
- Trained a customized YOLOv8 object-detection model for tennis footage, managing the full pipeline from manual data collection and labeling to dataset preparation, model training, and evaluation
- Combined player detection, ball tracking, and coordinate mapping to transform raw match footage into a birds-eye-view of on-court activity.

Team Captain, NCAA Division III Women's Tennis, CMS Student-Athlete - Claremont, CA *Aug 2024 - Present*

- Selected as team captain for the 2026–27 season, coordinating practices, mentoring teammates, and fostering team communication and accountability within a nationally competitive collegiate tennis program.

Selected Achievements: NCAA Division III Doubles National Champion (2026) • ITA Cup Division III Doubles Champion & Singles Finalist (2025) • ITA Cup Division III Singles Finalist (2026)

PROJECTS

Autonomous Underwater Vehicle *Aug 2025- May 2026*

Arduino, MatLab

- Designed and integrated multi-sensor circuitry for an autonomous underwater vehicle supporting ocean-sampling and environmental monitoring.
- Interfaced six sensor systems–including GPS, IMU, pressure, temperature, motor-current, and magnetometer sensors–with Arduino for onboard data acquisition.
- Developed MATLAB code to analyze sensor data for water-current mapping and magnetic-object detection.